INCH-POUND
MIL-PRF-39018/1J
9 August 2021
SUPERSEDING
MIL-PRF-39018/1H
20 February 2009

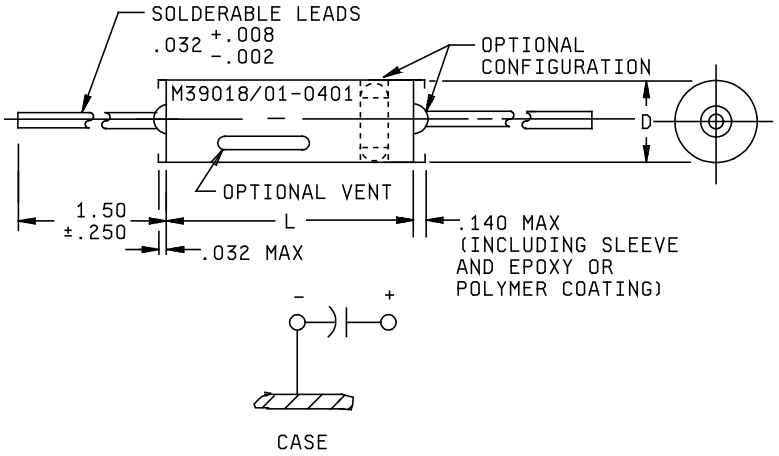
PERFORMANCE SPECIFICATION SHEET

# CAPACITOR, FIXED, ELECTROLYTIC (ALUMINUM OXIDE) (POLARIZED), NON-ESTABLISHED RELIABILITY AND ESTABLISHED RELIABILITY, STYLES CU12 (UNINSULATED), CU13 (INSULATED), AND CUR13 (INSULATED)

STYLE CU12 INACTIVE FOR DESIGN AFTER 23 MARCH 1972

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and MIL-PRF-39018.




| Case size | Dimensions | |
|---|---|---|
| | L ± .031 inch | D ± .015 inch (see note 4) |
| A0 | .938 | .281 |
| A1 | .938 | .375 |
| A2 | 1.125 | .375 |
| A3 | 1.375 | .375 |
| A4 | 1.625 | .375 |
| A5 | 2.188 | .375 |
| A6 | 2.688 | .375 |

| Inches | mm | Inches | mm |
|---|---|---|---|
| .002 | 0.05 | .375 | 9.53 |
| .008 | 0.20 | .938 | 23.83 |
| .015 | 0.38 | 1.125 | 28.57 |
| .031 | 0.79 | 1.375 | 34.92 |
| .032 | 0.81 | 1.50 | 38.1 |
| .140 | 3.56 | 1.625 | 41.27 |
| .250 | 6.35 | 2.188 | 55.58 |
| .281 | 7.14 | 2.688 | 68.28 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. These capacitors are not intended to be mounted by their leads.
4. The “D” dimensions for styles CU13 and CUR13 shall be not more than .008 inch (0.20 mm) larger than the specified bare case diameter.

FIGURE 1. Styles CU12, CU13, and CUR13 capacitors.

AMSC N/A
FSC 5910
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case:

Type: Tubular.

Material: Metal.

Leads: See figure 1.

DC rated voltage: See table I.

Reverse voltage: 3.0 volts maximum.

Capacitance (Cap.) value: See table I.

Cap. tolerance: See table I.

Failure rate level (CUR13 only): M, P, R, and S in accordance with MIL-PRF-39018.

Operating temperature range: -55°C to +85°C derated to +125°C (see table I).

Burn-in (CUR13 only): In accordance with MIL-PRF-39018.

DC leakage (DCL): See table I.

Equivalent series resistance (ESR): See table I.

Low temperature exposure:

DCL: See table I.

Cap.: See table I.

ESR: See table I.

Solderability: In accordance with MIL-PRF-39018.

Terminal strength: MIL-STD-202-211, test condition A (3 pounds) and test condition D (3 rotations).

Stability at reduced and high temperatures:

Step 1 (at +25°C):

DCL: See table I.

Cap.: See table I.

ESR: See table I.

Step 2 (at -55°C):

Impedance: See table I.

ΔCap.: Within ±30 percent of step 1 value.



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Step 3 (at +25°C):

DCL: See table I.

ΔCap.: Within ±5 percent of step 1 value.

ESR: See table I.

Step 4 (at +125°C):

DCL: See table I.

ΔCap.: Within ±25 percent of step 1 value.

ESR: See table I.

Step 5 (at +25°C):

DCL: See table I.

ΔCap.: Within ±5 percent of step 1 value.

ESR: See table I.

Life: MIL-STD-202-108.

For qualification: Condition F (2,000 hours) at +85°C.

DCL: See table I.

ΔCap.: Within ±15 percent of initial measured value.

ESR: Not more than 130 percent of initial requirement.

For quality conformance: Performance check, condition B (250 hours) at +85°C (non-ER).

DCL: See table I.

ΔCap.: Within ±10 percent of initial measured value.

ESR: See table I.

For quality conformance: Continuation test (1,750 hours) at +85°C (non-ER).

DCL: See table I.

ΔCap.: Within ±15 percent of initial measured value.

ESR: Not more than 130 percent of initial requirement.



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For quality conformance: 10,000 hours (ER) at +85°C.

DCL: At +25°C initial limit, at +85°C, 125 percent of initial limit.

ΔCap.: Within ±25 percent of initial measured value.

ESR: Not more than 200 percent of initial requirement.

Case insulation (CU13 and CUR13 only): In accordance with MIL-PRF-39018.

Vibration:

High frequency: In accordance with MIL-PRF-39018.

Random (style CUR13 only) (when applicable): In accordance with MIL-PRF-39018.

Thermal shock and immersion:

Thermal shock: MIL-STD-202-107, condition A, except in step 3, units shall be tested at applicable high temperature.

Immersion: MIL-STD-202-104, condition B (2 cycles, 30 minutes).

DCL: See table I.

ΔCap.: Within ±5 percent of initial measured value.

ESR: See table I.

Case insulation (CU13 and CUR13 only): Shall meet initial requirement.

Surge voltage:

DCL: See table I.

ΔCap.: Within ±6 percent of initial measured value.

ESR: See table I.

Moisture resistance: MIL-STD-202-106.

DCL: See table I.

ΔCap: Within ±6 percent of initial measured value.

ESR: See table I.

Case Insulation (CU13 and CUR13 only): Shall meet initial requirement.

High temperature exposure (500 hours at +125°C):

DCL: Not more than 200 percent of initial requirement for units less than 150 volts and 300 percent for units 150 volts and greater.

ΔCap.: Within ±10 percent of initial measured value.

ESR: Not more than 115 percent of initial requirement.



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Reverse voltage:

After 125 hours at +125°C with 1 +0 -.1 reverse volt applied to units rated at 7 V dc, and 3 +0 -.3 reverse volts applied to all others.

After additional 125 hours at +125°C with rated forward voltage applied:

DCL: Within 125 percent of initial requirement.

ΔCap.: Within the following percentages of the initial measured value:

7 to 30 V dc inclusive: +10, -25 percent.

50 to 350 V dc inclusive: ±10 percent.

ESR: See table I.

Penetration of solvents: In accordance with MIL-PRF-39018.

High temperature verification: In accordance with MIL-PRF-39018 at +125°C with derated voltage applied:

DCL: See table I.

ΔCap: Within 10 percent of initial measured value.

ESR: Not more than 130 percent of initial requirement.

Reverse voltage aging: In accordance with MIL-PRF-39018.

Marking: In accordance with MIL-PRF-39018.

Part or Identifying Number (PIN): M39018/01- (Dash number from table I). For style CUR13, parts with random vibration, the letter R shall be used in place of the dash (i.e., M39018/01R1046M).

Substitutability data: See table II.



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TABLE I. Styles CU12, CU13, and CUR13 capacitors.

| Cap | ESR | | Max. Imped. at -55°C and 120 Hz | DC leakage | | Max. ac ripple current 120 Hz at +85°C 1/ | Case size | Dash number | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Style CU12 | | | Style CU13 | | | Style CUR13 2/ | | |
| | | | | | | | | Capacitance tolerance | | | Capacitance tolerance | | | Capacitance tolerance | | |
| | +25°C | +85°C and +125°C | | +25° C | +85°C and +125°C | | | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 |
| µF | Ω | | Ω | µA | | mA | | Percent | | | Percent | | | Percent | | |
| Rated voltage: 7$V_{dc}$ at +85°C, 5$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 10$V_{dc}$ at +85°C, 7$V_{dc}$ at +125°C | | | | | | | | |
| 220 | 1.21 | 1.51 | 20 | 3 | 18 | 290 3/ | A1 | 0401 | --- | 0501 | 0601 | --- | 0701 | 1001- | --- | 1101- |
| 270 | .98 | 1.23 | 17 | 3 | 18 | 360 3/ | A2 | 0402 | --- | 0502 | 0602 | --- | 0702 | 1002- | --- | 1102- |
| 390 | .68 | .85 | 12 | 4 | 24 | 520 3/ | A3 | 0403 | --- | 0503 | 0603 | --- | 0703 | 1003- | --- | 1103- |
| 560 | .47 | .59 | 8 | 5 | 30 | 750 3/ | A4 | 0404 | --- | 0504 | 0604 | --- | 0704 | 1004- | --- | 1104- |
| 820 | .32 | .40 | 5.5 | 7 | 42 | 1100 3/ | A5 | 0405 | --- | 0505 | 0605 | --- | 0705 | 1005- | --- | 1105- |
| 1000 | .27 | .33 | 4.5 | 10 | 60 | 1350 3/ | A6 | 0406 | --- | 0506 | 0606 | --- | 0706 | 1006- | --- | 1106- |
| Rated voltage: 10$V_{dc}$ at +85°C, 7$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 15$V_{dc}$ at +85°C, 10$V_{dc}$ at +125°C | | | | | | | | |
| 100 | 2.65 | 3.32 | 38 | 2 | 12 | 285 3/ | A0 | 0407 | --- | 0507 | 0607 | --- | 0707 | 1007- | --- | 1107- |
| 180 | 1.47 | 1.84 | 25 | 3 | 18 | 438 3/ | A1 | 0408 | --- | 0508 | 0608 | --- | 0708 | 1008- | --- | 1108- |
| 220 | 1.21 | 1.51 | 20 | 3 | 18 | 527 3/ | A2 | 0409 | --- | 0509 | 0609 | --- | 0709 | 1009- | --- | 1109- |
| 330 | .80 | 1.00 | 14 | 4 | 24 | 696 3/ | A3 | 0410 | --- | 0510 | 0610 | --- | 0710 | 1010- | --- | 1110- |
| 470 | .56 | .71 | 9.6 | 5 | 30 | 892 3/ | A4 | 0411 | --- | 0511 | 0611 | --- | 0711 | 1011- | --- | 1111- |
| 680 | .39 | .49 | 6.6 | 7 | 42 | 1210 | A5 | 0412 | --- | 0512 | 0612 | --- | 0712 | 1012- | --- | 1112- |
| 820 | .32 | .40 | 5.5 | 10 | 60 | 1490 | A6 | 0413 | --- | 0513 | 0613 | --- | 0713 | 1013- | --- | 1113- |
| Rated voltage: 15$V_{dc}$ at +85°C, 10$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 20$V_{dc}$ at +85°C, 15$V_{dc}$ at +125°C | | | | | | | | |
| 68 | 3.90 | 4.86 | 66 | 2 | 12 | 181 3/ | A0 | 0414 | --- | 0514 | 0614 | --- | 0714 | 1014- | --- | 1114- |
| 82 | 3.23 | 4.04 | 55 | 2 | 12 | 219 3/ | A0 | 0415 | --- | 0515 | 0615 | --- | 0715 | 1015- | --- | 1115- |
| 150 | 1.77 | 2.21 | 30 | 3 | 18 | 400 | A1 | 0416 | --- | 0516 | 0616 | --- | 0716 | 1016- | --- | 1116- |
| 180 | 1.47 | 1.84 | 25 | 3 | 18 | 475 | A2 | 0417 | --- | 0517 | 0617 | --- | 0717 | 1017- | --- | 1117- |
| 270 | .98 | 1.23 | 17 | 4 | 24 | 628 | A3 | 0418 | --- | 0518 | 0618 | --- | 0718 | 1018- | --- | 1118- |
| 390 | .68 | .85 | 12 | 5 | 30 | 810 | A4 | 0419 | --- | 0519 | 0619 | --- | 0719 | 1019- | --- | 1119- |
| 560 | .47 | .59 | 8 | 7 | 42 | 1100 | A5 | 0420 | --- | 0520 | 0620 | --- | 0720 | 1020- | --- | 1120- |
| 680 | .39 | .49 | 6.6 | 10 | 60 | 1320 | A6 | 0421 | --- | 0521 | 0621 | --- | 0721 | 1021- | --- | 1121- |
| Rated voltage: 30$V_{dc}$ at +85°C, 20$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 40$V_{dc}$ at +85°C, 30$V_{dc}$ at +125°C | | | | | | | | |
| 33 | 8.04 | 10.05 | 91 | 2 | 12 | 164 | A0 | 0422 | --- | 0522 | 0622 | --- | 0722 | 1022- | --- | 1122- |
| 47 | 5.64 | 7.05 | 64 | 2 | 12 | 196 | A0 | 0423 | --- | 0523 | 0623 | --- | 0723 | 1023- | --- | 1123- |
| 82 | 3.23 | 4.04 | 36 | 3 | 18 | 297 | A1 | 0424 | --- | 0524 | 0624 | --- | 0724 | 1024- | --- | 1124- |
| 100 | 2.65 | 3.33 | 30 | 3 | 18 | 353 | A2 | 0425 | --- | 0525 | 0625 | --- | 0725 | 1025- | --- | 1125- |
| 120 | 2.21 | 2.76 | 25 | 4 | 24 | 419 | A3 | 0426 | --- | 0526 | 0626 | --- | 0726 | 1026- | --- | 1126- |
| 150 | 1.77 | 2.21 | 20 | 4 | 24 | 470 | A3 | 0427 | --- | 0527 | 0627 | --- | 0727 | 1027- | --- | 1127- |
| 220 | 1.20 | 1.51 | 14 | 5 | 30 | 607 | A4 | 0428 | --- | 0528 | 0628 | --- | 0728 | 1028- | --- | 1128- |
| 330 | .80 | 1.00 | 9.1 | 7 | 42 | 848 | A5 | 0429 | --- | 0529 | 0629 | --- | 0729 | 1029- | --- | 1129- |
| 390 | .68 | .85 | 7.7 | 10 | 60 | 1000 | A6 | 0430 | --- | 0530 | 0630 | --- | 0730 | 1030- | --- | 1130- |
| Rated voltage: 50$V_{dc}$ at +85°C, 40$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 60$V_{dc}$ at +85°C, 50$V_{dc}$ at +125°C | | | | | | | | |
| 22 | 9.04 | 15.07 | 140 | 2 | 12 | 134 | A0 | 0431 | --- | 0531 | 0631 | --- | 0731 | 1031- | --- | 1131- |
| 33 | 5.03 | 10.05 | 91 | 3 | 18 | 188 | A1 | 0432 | --- | 0532 | 0632 | --- | 0732 | 1032- | --- | 1132- |
| 47 | 4.23 | 7.05 | 64 | 3 | 18 | 243 | A2 | 0433 | --- | 0533 | 0633 | --- | 0733 | 1033- | --- | 1133- |
| 56 | 3.55 | 5.92 | 54 | 4 | 24 | 286 | A3 | 0434 | --- | 0534 | 0634 | --- | 0734 | 1034- | --- | 1134- |
| 68 | 2.92 | 4.88 | 44 | 4 | 24 | 315 | A3 | 0435 | --- | 0535 | 0635 | --- | 0735 | 1035- | --- | 1135- |
| 100 | 1.99 | 3.32 | 30 | 5 | 30 | 410 | A4 | 0436 | --- | 0536 | 0636 | --- | 0736 | 1036- | --- | 1136- |
| 150 | 1.33 | 2.21 | 20 | 7 | 42 | 571 | A5 | 0437 | --- | 0537 | 0637 | --- | 0737 | 1037- | --- | 1137- |
| 180 | 1.11 | 1.84 | 17 | 10 | 60 | 680 | A6 | 0438 | --- | 0538 | 0638 | --- | 0738 | 1038- | --- | 1138- |
| Rated voltage: 75$V_{dc}$ at +85°C, 60$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 90$V_{dc}$ at +85°C, 75$V_{dc}$ at +125°C | | | | | | | | |
| 12 | 16.58 | 27.63 | 250 | 2 | 12 | 96 | A0 | 0439 | --- | 0539 | 0639 | --- | 0739 | 1039- | --- | 1139- |
| 22 | 9.04 | 15.07 | 140 | 3 | 18 | 153 | A1 | 0440 | --- | 0540 | 0640 | --- | 0740 | 1040- | --- | 1140- |
| 33 | 6.03 | 10.05 | 91 | 3 | 18 | 204 | A2 | 0441 | --- | 0541 | 0641 | --- | 0741 | 1041- | --- | 1141- |
| 47 | 4.27 | 7.05 | 64 | 4 | 24 | 262 | A3 | 0442 | --- | 0542 | 0642 | --- | 0742 | 1042- | --- | 1142- |
| 68 | 2.92 | 4.88 | 44 | 5 | 30 | 337 | A4 | 0443 | --- | 0543 | 0643 | --- | 0743 | 1043- | --- | 1143- |
| 100 | 1.99 | 3.32 | 30 | 7 | 42 | 506 | A5 | 0444 | --- | 0544 | 0644 | --- | 0744 | 1044- | --- | 1144- |
| 120 | 1.65 | 2.76 | 25 | 10 | 60 | 555 | A6 | 0445 | --- | 0545 | 0645 | --- | 0745 | 1045- | --- | 1145- |

See footnotes at end of table.



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TABLE I. Styles CU12, CU13, and CUR13 capacitors - Continued.

| Cap | ESR | | Max. Imped. at -55°C and 120 Hz | DC leakage | | Max. ac ripple current 120 Hz at +85°C 1/ | Case size | Dash number | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Style CU12 | | | Style CU13 | | | Style CUR13 2/ | | |
| | | | | | | | | Capacitance tolerance | | | Capacitance tolerance | | | Capacitance tolerance | | |
| | +25°C | +85°C and +125°C | | +25° C | +85°C and +125°C | | | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 |
| µF | Ω | | Ω | µA | | mA | | Percent | | | Percent | | | Percent | | |
| Rated voltage: 100$V_{dc}$ at +85°C, 75$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 125$V_{dc}$ at +85°C, 100$V_{dc}$ at +125°C | | | | | | | | |
| 8.2 | 24.26 | 40.44 | 360 | 2 | 12 | 82 | A0 | 0446 | --- | 0546 | 0646 | --- | 0746 | 1046- | --- | 1146- |
| 12 | 16.54 | 27.63 | 250 | 3 | 18 | 109 | A1 | 0447 | --- | 0547 | 0647 | --- | 0747 | 1047- | --- | 1147- |
| 15 | 13.26 | 22.10 | 200 | 3 | 18 | 137 | A2 | 0448 | --- | 0548 | 0648 | --- | 0748 | 1048- | --- | 1148- |
| 22 | 9.04 | 15.07 | 140 | 4 | 24 | 179 | A3 | 0449 | --- | 0549 | 0649 | --- | 0749 | 1049- | --- | 1149- |
| 33 | 6.03 | 10.05 | 91 | 5 | 30 | 235 | A4 | 0450 | --- | 0550 | 0650 | --- | 0750 | 1050- | --- | 1150- |
| 47 | 4.23 | 7.05 | 64 | 7 | 42 | 319 | A5 | 0451 | --- | 0551 | 0651 | --- | 0751 | 1051- | --- | 1151- |
| 68 | 2.92 | 4.88 | 44 | 10 | 60 | 417 | A6 | 0452 | --- | 0552 | 0652 | --- | 0752 | 1052- | --- | 1152- |
| Rated voltage: 150$V_{dc}$ at +85°C, 100$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 175$V_{dc}$ at +85°C, 125$V_{dc}$ at +125°C | | | | | | | | |
| 4.7 | 28.22 | 56.44 | 640 | 2 | 12 | 69 | A0 | 0453 | 0553 | --- | 0653 | 0753 | --- | 1053- | 1153- | --- |
| 5.6 | 23.68 | 47.37 | 540 | 2 | 12 | 75 | A0 | 0454 | 0554 | --- | 0654 | 0754 | --- | 1054- | 1154- | --- |
| 8.2 | 10.17 | 32.35 | 360 | 3 | 18 | 105 | A1 | 0455 | 0555 | --- | 0655 | 0755 | --- | 1055- | 1155- | --- |
| 12 | 11.05 | 22.10 | 250 | 3 | 18 | 137 | A2 | 0456 | 0556 | --- | 0656 | 0756 | --- | 1056- | 1156- | --- |
| 18 | 7.37 | 14.74 | 170 | 4 | 24 | 182 | A3 | 0457 | 0557 | --- | 0657 | 0757 | --- | 1057- | 1157- | --- |
| 22 | 6.03 | 12.06 | 140 | 5 | 30 | 214 | A4 | 0458 | 0558 | --- | 0658 | 0758 | --- | 1058- | 1158- | --- |
| 33 | 4.02 | 8.04 | 91 | 7 | 42 | 300 | A5 | 0459 | 0559 | --- | 0659 | 0759 | --- | 1059- | 1159- | --- |
| 56 | 2.37 | 4.74 | 54 | 10 | 60 | 423 | A6 | 0460 | 0560 | --- | 0660 | 0760 | --- | 1060- | 1160- | --- |
| Rated voltage: 200$V_{dc}$ at +85°C, 150$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 225$V_{dc}$ at +85°C, 175$V_{dc}$ at +125°C | | | | | | | | |
| 3.3 | 40.30 | 80.38 | 770 | 4 | 24 | 63 | A0 | 0490 | 0590 | --- | 0690 | 0790 | --- | 1061- | 1161- | --- |
| 3.9 | 34.01 | 68.02 | 500 | 4 | 24 | 68 | A0 | 0461 | 0561 | --- | 0661 | 0761 | --- | 1089- | 1189- | --- |
| 5.6 | 23.68 | 47.37 | 333 | 6 | 36 | 86 | A1 | 0462 | 0562 | --- | 0662 | 0762 | --- | 1062- | 1162- | --- |
| 8.2 | 16.17 | 32.35 | 250 | 6 | 36 | 113 | A2 | 0463 | 0563 | --- | 0663 | 0763 | --- | 1063- | 1163- | --- |
| 12 | 11.05 | 22.10 | 154 | 8 | 48 | 148 | A3 | 0464 | 0564 | --- | 0664 | 0764 | --- | 1064- | 1164- | --- |
| 15 | 8.84 | 17.68 | 118 | 10 | 60 | 177 | A4 | 0465 | 0565 | --- | 0665 | 0765 | --- | 1065- | 1165- | --- |
| 27 | 4.91 | 9.82 | 77 | 14 | 84 | 271 | A5 | 0466 | 0566 | --- | 0666 | 0766 | --- | 1066- | 1166- | --- |
| 39 | 3.40 | 6.80 | 51 | 20 | 120 | 353 | A6 | 0467 | 0567 | --- | 0667 | 0767 | --- | 1067- | 1167- | --- |
| Rated voltage: 250$V_{dc}$ at +85°C, 200$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 275$V_{dc}$ at +85°C, 225$V_{dc}$ at +125°C | | | | | | | | |
| 2.2 | 60.45 | 120.57 | 910 | 4 | 24 | 58 | A0 | 0491 | 0591 | --- | 0691 | 0791 | --- | 1068- | 1168- | --- |
| 3.3 | 40.19 | 80.38 | 910 | 4 | 24 | 70 | A0 | 0468 | 0568 | --- | 0668 | 0768 | --- | 1090- | 1190- | --- |
| 5.6 | 23.68 | 47.37 | 540 | 6 | 36 | 86 | A1 | 0469 | 0569 | --- | 0669 | 0769 | --- | 1069- | 1169- | --- |
| 6.8 | 19.50 | 39.01 | 440 | 6 | 36 | 103 | A2 | 0470 | 0570 | --- | 0670 | 0770 | --- | 1070- | 1170- | --- |
| 10 | 13.26 | 26.53 | 300 | 8 | 48 | 125 | A3 | 0471 | 0571 | --- | 0671 | 0771 | --- | 1071- | 1171- | --- |
| 12 | 11.05 | 22.10 | 250 | 10 | 60 | 159 | A4 | 0472 | 0572 | --- | 0672 | 0772 | --- | 1072- | 1172- | --- |
| 22 | 6.03 | 12.06 | 140 | 14 | 84 | 245 | A5 | 0473 | 0573 | --- | 0673 | 0773 | --- | 1073- | 1173- | --- |
| 27 | 4.91 | 9.82 | 110 | 20 | 120 | 294 | A6 | 0474 | 0574 | --- | 0674 | 0774 | --- | 1074- | 1174- | --- |
| Rated voltage: 300$V_{dc}$ at +85°C, 225$V_{dc}$ at +125°C | | | | | | | | Surge voltage: 325$V_{dc}$ at +85°C, 250$V_{dc}$ at +125°C | | | | | | | | |
| 1.5 | 132.66 | 221.05 | 1300 | 16 | 96 | 42 | A0 | 0492 | 0592 | --- | 0692 | 0792 | --- | 1075- | 1175- | --- |
| 2.2 | 90.43 | 150.71 | 1300 | 16 | 96 | 51 | A0 | 0475 | 0575 | --- | 0675 | 0775 | --- | 1091- | 1191- | --- |
| 3.9 | 51.01 | 85.02 | 640 | 20 | 120 | 71 | A1 | 0493 | 0593 | --- | 0693 | 0793 | --- | 1076- | 1176- | --- |
| 4.7 | 42.33 | 70.55 | 640 | 20 | 120 | 78 | A1 | 0476 | 0576 | --- | 0676 | 0776 | --- | 1092- | 1192- | --- |
| 5.6 | 35.53 | 59.21 | 440 | 22 | 132 | 84 | A2 | 0489 | 0589 | --- | 0689 | 0789 | --- | 1077- | 1177- | --- |
| 8.2 | 24.26 | 40.44 | 370 | 24 | 144 | 110 | A3 | 0478 | 0578 | --- | 0678 | 0778 | --- | 1078- | 1178- | --- |
| 10 | 19.89 | 33.16 | 300 | 26 | 156 | 130 | A4 | 0479 | 0579 | --- | 0679 | 0779 | --- | 1079- | 1179- | --- |
| 16 | 12.43 | 20.72 | 200 | 30 | 180 | 185 | A5 | 0480 | 0580 | --- | 0680 | 0780 | --- | 1080- | 1180- | --- |
| 18 | 11.05 | 18.42 | 170 | 36 | 216 | 217 | A6 | 0481 | 0581 | --- | 0681 | 0781 | --- | 1081- | 1181- | --- |

See footnote at end of table.



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TABLE I. Styles CU12, CU13, and CUR13 capacitors - Continued.

| Cap | ESR | | Max. Imped. at -55°C and 120 Hz | DC leakage | | Max. ac ripple current 120 Hz at +85°C 1/ | Case size | Dash number | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Style CU12 | | | Style CU13 | | | Style CUR13 2/ | | |
| | | | | | | | | Capacitance tolerance | | | Capacitance tolerance | | | Capacitance tolerance | | |
| | +25°C | +85°C and +125°C | | +25° C | +85°C and +125°C | | | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 |
| µF | Ω | | Ω | µA | | mA | | Percent | | | Percent | | | Percent | | |
| Rated voltage: 350$V_{dc}$ at +85°C, 275$V_{dc}$ at +125°C | | | | | | | Surge voltage: 375$V_{dc}$ at +85°C, 300$V_{dc}$ at +125°C | | | | | | | | | |
| 1.0 | 199.00 | 331.57 | 2000 | 20 | 120 | 35 | A0 | 0494 | 0594 | --- | 0694 | 0794 | --- | 1082- | 1182- | --- |
| 3.3 | 60.29 | 100.42 | 910 | 26 | 156 | 59 | A1 | 0483 | 0583 | --- | 0683 | 0783 | --- | 1083- | 1183 | --- |
| 3.9 | 51.01 | 85.02 | 770 | 28 | 168 | 70 | A2 | 0484 | 0584 | --- | 0684 | 0784 | --- | 1084- | 1184- | --- |
| 5.6 | 35.53 | 59.21 | 540 | 32 | 192 | 90 | A3 | 0485 | 0585 | --- | 0685 | 0785 | --- | 1085- | 1185- | --- |
| 6.8 | 29.26 | 48.76 | 440 | 34 | 204 | 107 | A4 | 0486 | 0586 | --- | 0686 | 0786 | --- | 1086- | 1186- | --- |
| 10 | 19.89 | 33.16 | 300 | 38 | 228 | 148 | A5 | 0487 | 0587 | --- | 0687 | 0787 | --- | 1087- | 1187- | --- |
| 12 | 16.58 | 27.63 | 250 | 42 | 252 | 174 | A6 | 0488 | 0588 | --- | 0688 | 0788 | --- | 1088- | 1188- | --- |

1/ The ripple current values shown are representative values not supported by test data.
2/ The last letter in dash number will indicate FR level symbol ("M", "P", "R", or "S"). For style CUR13 with random vibration, the letter "R" will be used in place of the dash between the spec sheet number and the dash number (i.e., M39018/01R1046M).
3/ Maximum current due to voltage limitations.

TABLE II. Substitutability data.

| Dash number in MIL-PRF-39018/1A through /1J 1/ | | | | | | Dash number MIL-PRF-39018/1 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Style CU12 | | | Style CU13 | | | Style CU12 | | | Style CU13 | | |
| Capacitance tolerance | | | Capacitance tolerance | | | Capacitance tolerance | | | Capacitance tolerance | | |
| -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 |
| 0401 | --- | 0501 | 0601 | --- | 0701 | 0005 | --- | 0006 | 0007 | --- | 0008 |
| 0402 | --- | 0502 | 0602 | --- | 0702 | 0009 | --- | 0010 | 0011 | --- | 0012 |
| 0403 | --- | 0503 | 0603 | --- | 0703 | 0013 | --- | 0014 | 0015 | --- | 0016 |
| 0404 | --- | 0504 | 0604 | --- | 0704 | 0017 | --- | 0018 | 0019 | --- | 0020 |
| 0405 | --- | 0505 | 0605 | --- | 0705 | 0021 | --- | 0022 | 0023 | --- | 0024 |
| 0406 | --- | 0506 | 0606 | --- | 0706 | 0025 | --- | 0026 | 0027 | --- | 0028 |
| --- | --- | --- | --- | --- | --- | 0001 | --- | 0002 | 0003 | --- | 0004 |
| 0407 | --- | 0507 | 0607 | --- | 0707 | 0029 | --- | 0030 | 0031 | --- | 0032 |
| 0408 | --- | 0508 | 0608 | --- | 0708 | 0033 | --- | 0034 | 0035 | --- | 0036 |
| 0409 | --- | 0509 | 0609 | --- | 0709 | 0037 | --- | 0038 | 0039 | --- | 0040 |
| 0410 | --- | 0510 | 0610 | --- | 0710 | 0041 | --- | 0042 | 0043 | --- | 0044 |
| 0411 | --- | 0511 | 0611 | --- | 0711 | 0045 | --- | 0046 | 0047 | --- | 0048 |
| 0412 | --- | 0512 | 0612 | --- | 0712 | 0049 | --- | 0050 | 0051 | --- | 0052 |
| 0413 | --- | 0513 | 0613 | --- | 0713 | 0053 | --- | 0054 | 0055 | --- | 0056 |
| 0414 | --- | 0514 | 0614 | --- | 0714 | 0057 | --- | 0058 | 0059 | --- | 0060 |
| 0415 | --- | 0515 | 0615 | --- | 0715 | 0061 | --- | 0062 | 0063 | --- | 0064 |
| 0416 | --- | 0516 | 0616 | --- | 0716 | 0065 | --- | 0066 | 0067 | --- | 0068 |
| 0417 | --- | 0517 | 0617 | --- | 0717 | 0069 | --- | 0070 | 0071 | --- | 0072 |
| 0418 | --- | 0518 | 0618 | --- | 0718 | 0073 | --- | 0074 | 0075 | --- | 0076 |
| 0419 | --- | 0519 | 0619 | --- | 0719 | 0077 | --- | 0078 | 0079 | --- | 0080 |
| 0420 | --- | 0520 | 0620 | --- | 0720 | 0081 | --- | 0082 | 0083 | --- | 0084 |
| 0421 | --- | 0521 | 0621 | --- | 0721 | 0085 | --- | 0086 | 0087 | --- | 0088 |
| 0422 | --- | 0522 | 0622 | --- | 0722 | 0089 | --- | 0090 | 0091 | --- | 0092 |
| 0423 | --- | 0523 | 0623 | --- | 0723 | 0093 | --- | 0094 | 0095 | --- | 0096 |
| 0424 | --- | 0524 | 0624 | --- | 0724 | 0097 | --- | 0098 | 0099 | --- | 0100 |
| 0425 | --- | 0525 | 0625 | --- | 0725 | 0101 | --- | 0102 | 0103 | --- | 0104 |
| 0426 | --- | 0526 | 0626 | --- | 0726 | 0105 | --- | 0106 | 0107 | --- | 0108 |
| 0427 | --- | 0527 | 0627 | --- | 0727 | 0109 | --- | 0110 | 0111 | --- | 0112 |

See footnote at end of table.



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TABLE II. Substitutability data - Continued.

| Dash number in MIL-PRF-39018/1A through /1J 1/ | | | | | | Dash number MIL-PRF-39018/1 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Style CU12 | | | Style CU13 | | | Style CU12 | | | Style CU13 | | |
| Capacitance tolerance | | | Capacitance tolerance | | | Capacitance tolerance | | | Capacitance tolerance | | |
| -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 | -10 +30 | -10 +50 | -10 +75 |
| 0428 | --- | 0528 | 0628 | --- | 0728 | 0113 | --- | 0114 | 0115 | --- | 0116 |
| 0429 | --- | 0529 | 0629 | --- | 0729 | 0117 | --- | 0118 | 0119 | --- | 0120 |
| 0430 | --- | 0530 | 0630 | --- | 0730 | 0121 | --- | 0122 | 0123 | --- | 0124 |
| 0431 | --- | 0531 | 0631 | --- | 0731 | 0125 | --- | 0126 | 0127 | --- | 0128 |
| 0432 | --- | 0532 | 0632 | --- | 0732 | 0129 | --- | 0130 | 0131 | --- | 0132 |
| 0433 | --- | 0533 | 0633 | --- | 0733 | 0133 | --- | 0134 | 0135 | --- | 0136 |
| 0434 | --- | 0534 | 0634 | --- | 0734 | 0137 | --- | 0138 | 0139 | --- | 0140 |
| 0435 | --- | 0535 | 0635 | --- | 0735 | 0141 | --- | 0142 | 0143 | --- | 0144 |
| 0436 | --- | 0536 | 0636 | --- | 0736 | 0145 | --- | 0146 | 0147 | --- | 0148 |
| 0437 | --- | 0537 | 0637 | --- | 0737 | 0149 | --- | 0150 | 0151 | --- | 0152 |
| 0438 | --- | 0538 | 0638 | --- | 0738 | 0153 | --- | 0154 | 0155 | --- | 0156 |
| 0439 | --- | 0539 | 0639 | --- | 0739 | 0157 | --- | 0158 | 0159 | --- | 0160 |
| 0440 | --- | 0540 | 0640 | --- | 0740 | 0161 | --- | 0162 | 0163 | --- | 0164 |
| 0441 | --- | 0541 | 0641 | --- | 0741 | 0165 | --- | 0166 | 0167 | --- | 0168 |
| 0442 | --- | 0542 | 0642 | --- | 0742 | 0169 | --- | 0170 | 0171 | --- | 0172 |
| 0443 | --- | 0543 | 0643 | --- | 0743 | 0173 | --- | 0174 | 0175 | --- | 0176 |
| 0444 | --- | 0544 | 0644 | --- | 0744 | 0177 | --- | 0178 | 0179 | --- | 0180 |
| 0445 | --- | 0545 | 0645 | --- | 0745 | 0181 | --- | 0182 | 0183 | --- | 0184 |
| 0446 | 0546 | --- | 0646 | 0746 | --- | 0185 | 0186 | --- | 0187 | 0188 | --- |
| 0447 | 0547 | --- | 0647 | 0747 | --- | 0189 | 0190 | --- | 0191 | 0192 | --- |
| 0448 | 0548 | --- | 0648 | 0748 | --- | 0193 | 0194 | --- | 0195 | 0196 | --- |
| 0449 | 0549 | --- | 0649 | 0749 | --- | 0197 | 0198 | --- | 0199 | 0200 | --- |
| 0450 | 0550 | --- | 0650 | 0750 | --- | 0201 | 0202 | --- | 0203 | 0204 | --- |
| 0451 | 0551 | --- | 0651 | 0751 | --- | 0205 | 0206 | --- | 0207 | 0208 | --- |
| 0452 | 0552 | --- | 0652 | 0752 | --- | 0209 | 0210 | --- | 0211 | 0212 | --- |
| 0453 | 0553 | --- | 0653 | 0753 | --- | 0213 | 0214 | --- | 0215 | 0216 | --- |
| 0454 | 0554 | --- | 0654 | 0754 | --- | 0217 | 0218 | --- | 0219 | 0220 | --- |
| 0455 | 0555 | --- | 0655 | 0755 | --- | 0221 | 0222 | --- | 0223 | 0224 | --- |
| 0456 | 0556 | --- | 0656 | 0756 | --- | 0225 | 0226 | --- | 0227 | 0228 | --- |
| 0457 | 0557 | --- | 0657 | 0757 | --- | 0229 | 0230 | --- | 0231 | 0232 | --- |
| 0458 | 0558 | --- | 0658 | 0758 | --- | 0233 | 0234 | --- | 0235 | 0236 | --- |
| 0459 | 0559 | --- | 0659 | 0759 | --- | 0237 | 0238 | --- | 0239 | 0240 | --- |
| 0460 | 0560 | --- | 0660 | 0760 | --- | 0241 | 0242 | --- | 0243 | 0244 | --- |
| 0461 | 0561 | --- | 0661 | 0761 | --- | --- | --- | --- | --- | --- | --- |
| 0462 | 0562 | --- | 0662 | 0762 | --- | --- | --- | --- | --- | --- | --- |
| 0463 | 0563 | --- | 0663 | 0763 | --- | --- | --- | --- | --- | --- | --- |
| 0464 | 0564 | --- | 0664 | 0764 | --- | --- | --- | --- | --- | --- | --- |
| 0465 | 0565 | --- | 0665 | 0765 | --- | --- | --- | --- | --- | --- | --- |
| 0466 | 0566 | --- | 0666 | 0766 | --- | --- | --- | --- | --- | --- | --- |
| 0467 | 0567 | --- | 0667 | 0767 | --- | 0245 | 0246 | --- | 0247 | 0248 | --- |
| 0468 | 0568 | --- | 0668 | 0768 | --- | 0249 | 0250 | --- | 0251 | 0252 | --- |
| 0469 | 0569 | --- | 0669 | 0769 | --- | 0253 | 0254 | --- | 0255 | 0256 | --- |
| 0470 | 0570 | --- | 0670 | 0770 | --- | 0257 | 0258 | --- | 0259 | 0260 | --- |
| 0471 | 0571 | --- | 0671 | 0771 | --- | 0261 | 0262 | --- | 0263 | 0264 | --- |
| 0472 | 0572 | --- | 0672 | 0772 | --- | 0265 | 0266 | --- | 0267 | 0268 | --- |
| 0473 | 0573 | --- | 0673 | 0773 | --- | 0269 | 0270 | --- | 0271 | 0272 | --- |
| 0474 | 0574 | --- | 0674 | 0774 | --- | 0273 | 0274 | --- | 0275 | 0276 | --- |

1/ The following dash numbers have been canceled with no replacement items recommended:

| | | | |
|---|---|---|---|
| 0477 | 0577 | 0677 | 0777 |
| 0482 | 0582 | 0682 | 0782 |



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APPLICATION NOTE:

Ripple current: The maximum permissible ripple current for these capacitors can be determined from table III. The appropriate multiplier from this table should be used where operation at other frequencies or temperatures is desired. Operation of these capacitors outside the limits of 60 to 100,000 Hz is not recommended.

TABLE III. Ripple current multipliers.

| With temperature | +25°C | +45°C | +65°C | +85°C | +125°C |
|---|---|---|---|---|---|
| | 1.4 | 1.4 | 1.2 | 1.0 | 0.3 |
| With frequency | | 60 Hz | 120 Hz | 400 Hz | 1 K through |
| | 0- 60 volts | .85 | 1.0 | 1.1 | 1.15 |
| | 61-200 | .83 | 1.0 | 1.15 | 1.2 |
| | 201-350 | .80 | 1.0 | 1.3 | 1.4 |

Referenced documents: In addition to MIL-PRF-39018, this document references the following:

MIL-STD-202-104
MIL-STD-202-106
MIL-STD-202-107
MIL-STD-202-108
MIL-STD-202-211

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2021-019)

Review activities:
Navy - AS, MC, OS, SH
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